LITERATURE REVIEW

Deep Learning in the Sequence Space

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ANALYTICAL ABSTRACT

Dynamic stochastic general equilibrium models with heterogeneous agents are hard to solve globally because the aggregate state includes cross-sectional distributions. The paper replaces that state by a truncated history of exogenous aggregate shocks, which is an approximate sufficient statistic when the economy is ergodic, and trains deep neural networks to map the history to equilibrium objects by minimising equilibrium-condition residuals on simulated paths. A second contribution is a shape-preserving operator architecture: the network outputs an idiosyncratic policy function on a grid, built from positive increments so that it is monotone, concave and consistent with the borrowing limit by construction, which in turn permits the endogenous grid method to generate supervised targets. Three economies of rising difficulty are solved: a stochastic growth model, a 72-cohort overlapping generations economy with disasters, and a model with heterogeneous households and heterogeneous firms under uncertainty shocks. Mean relative residuals stay below 0.1 percent in the first two models and below 0.2 percent in the third. The weakness is validation: sufficiency of the truncation is proven only in a special case, and the hardest model has no external benchmark.

1 Introduction

1.1 Paper identification and classification

The paper is a computational and methodological contribution in macroeconomics, with JEL codes C61, C63, C68, D52 and E32. Its field is quantitative macroeconomics and its subfield is the global numerical solution of heterogeneous agent models with aggregate uncertainty, in the tradition of Krusell and Smith (1998), Reiter (2009) and Auclert, Bardoczy, Rognlie and Straub (2021). The version read in full is the arXiv HTML rendering of version 1 dated 16 September 2025. A PDF could not be retrieved in this session, so no copy of the paper is stored beside this review. Footnote text was not recoverable from the rendering, therefore footnote content is not evaluated here.

1.2 Executive understanding

WHAT IS THE PAPER TRYING TO DISCOVER?

The paper asks whether the infinite-dimensional endogenous aggregate state of a dynamic economy can be replaced by a finite window of past exogenous shocks without losing accuracy, and whether a neural network trained only on equilibrium residuals can then deliver a global solution in economies that conventional methods cannot handle.

WHY DOES THE QUESTION MATTER?

Heterogeneous agent models with aggregate risk are the workhorse for studying distributional consequences of business cycles, uncertainty shocks and disasters. Local methods based on sequence-space Jacobians are fast but impose certainty equivalence in aggregates, so they cannot capture risk premia, occasionally binding constraints or large shocks. Global methods exist but suffer from the curse of dimensionality in the distribution.

WHAT WAS KNOWN BEFORE, AND WHAT IS MISSING?

Krusell and Smith (1998) summarise the distribution by a few moments. Chien, Cole and Lustig (2011) and Lee (2025) already approximate the state by shock histories, but the first struggles with persistence and the second scales poorly with the number of aggregate shock values. Azinovic, Gaegauf and Scheidegger (2022) and Maliar, Maliar and Winant (2021) train networks on residuals using the endogenous state as input. Missing was a general, simulation-based global method that handles continuous and discrete shocks, regimes and rich cross-sectional heterogeneity with a stable training loop.

WHAT DOES THE PAPER PROPOSE, HOW DOES IT INVESTIGATE, AND WHAT DOES IT FIND?

The proposal combines three ingredients: truncated shock histories as network inputs, operator learning with shape constraints, and the endogenous grid method for targets. It is investigated by

solving three models and reporting residual distributions along simulated ergodic paths, plus a comparison with a grid solution in the simplest model. Residuals are small in all cases (Table 2).

2 Research Problem and Hypothesis

2.1 Research question

PAPER The paper asks whether, for ergodic dynamic economies, a neural network that takes the last T aggregate shock realisations as its only input can approximate the functional rational expectations equilibrium to within a fraction of one percent in relative equilibrium-condition error, including in economies with two endogenous distributions, where the truncation error is controlled by the contraction rate of the economy.

2.2 Hypotheses

INFERENCE The paper states no formal hypotheses; the following are reconstructed by the reviewer from its stated contributions.

- H1, sufficiency: for $|\rho| < 1$ and decaying dependence on initial conditions, the truncated history of shocks is an approximately sufficient statistic of the endogenous aggregate state, with error vanishing geometrically in T.
- H2, trainability: because the distribution of histories is exogenous and stationary, training avoids the feedback between the policy update and the distribution of training inputs that affects state-space methods.
- H3, shape preservation: encoding monotonicity, concavity and the borrowing limit in the output layer restricts search to economically admissible policies and enables supervised targets through the endogenous grid method.
- H4, scalability: the method solves models with 72 cohorts or with firm and household distributions that are out of reach of existing global methods.

2.3 Existing literature and research gap

PAPER The authors position the work against three literatures: local sequence-space linearisation (Boppart, Krusell and Mitman 2018; Auclert et al. 2021), shock-history global methods (Chien et al. 2011; Lee 2025), and deep-learning equilibrium solvers (Duffy and McNelis 2001; Azinovic et al. 2022; Maliar et al. 2021; Azinovic and Zemlicka 2024; Zhong 2023). The gap they identify is generality: a global method that is parallelisable and not tied to few discrete shocks.

CRITIQUE The gap is real, but the comparison is made only in words. No run-time, accuracy or memory comparison with Lee (2025), with state-space DEQN, or with Krusell and Smith type algorithms is reported for any of the three models.

2.4 Claimed contribution

CRITIQUE Four contributions are claimed. The sequence-space network and the shape-preserving architecture are demonstrated by construction, and their accuracy is demonstrated by residuals. The claim that the paper is the first to solve globally a model with incomplete markets and firm heterogeneity is a priority claim that this review cannot verify. The implicit claim that the sequence-space representation is superior to the state-space one is not tested, since no state-space baseline is trained on the same problems.

3 Theoretical Framework and Mechanism

3.1 The stochastic growth model and the logic of sufficiency

PAPER The testbed is the Brock and Mirman (1972) economy. Output is produced with

$$Y_t = A_t K_t^{\alpha} L^{1-\alpha}, \qquad \log A_t = \rho_A \log A_{t-1} + \sigma_A \varepsilon_t^A, \quad \varepsilon_t^A \sim \mathcal{N}(0,1), \tag{1}$$

- Y_t, K_t, L Output, capital stock and labour, with $L = 1$ supplied inelastically.
- A_t Stochastic total factor productivity; ρ_A is its persistence and σ_A the innovation volatility.
- α Capital elasticity of output, set to $1/3$.

and factor prices are the marginal products $r_t^K = \alpha A_t K_t^{\alpha-1} L^{\alpha}$ and $w_t = (1-\alpha) A_t K_t^{\alpha} L^{-\alpha}$. The representative household solves $\max \mathbb{E} \sum_t \beta^t u(C_t)$ subject to $C_t = L w_t + (1 - \delta + r_t^K) K_t - K_{t+1}$. The Euler equation, and its rearrangement in units of relative consumption error, are

$$u'(C_t) = \mathbb{E}\left[\beta\, u'(C_{t+1})\,(1 - \delta + r_{t+1}^K)\right], \tag{2}$$

$$0 = (u')^{-1}\Big(\mathbb{E}\left[\beta\, u'(C_{t+1})(1 - \delta + r_{t+1}^K)\right]\Big)\, C_t^{-1} - 1. \tag{3}$$

Equation (2) equates the marginal utility cost of saving one unit with the expected discounted marginal utility of its gross return. Dividing the consumption implied by the right side by actual consumption, as in (3), yields a dimensionless residual that equals the percentage by which the agent would change consumption if the Euler equation were imposed exactly. This is why the paper reports errors as relative consumption errors.

The state-space solution is $K_{t+1} = \pi(A_t, K_t)$. The sequence-space proposal is to learn instead $\mathcal{N}_\rho(\mathbf{X}_t^{seq,T}) \approx K_{t+1}$ with $\mathbf{X}_t^{seq,T} = [A_{t-T+1}, \dots, A_t]$ or the corresponding innovations. For this to be possible the history must determine K_t up to a vanishing error. The paper proves this only for $\delta = 1$ and $u = \log$, where the policy is $K_{t+1} = \alpha\beta A_t K_t^{\alpha}$. This closed form can be verified from (2): with $C_t = (1 - \alpha\beta) A_t K_t^{\alpha}$ one has $\beta\,\alpha A_{t+1} K_{t+1}^{\alpha-1} / C_{t+1} = \alpha\beta / ((1-\alpha\beta) K_{t+1})$, and equating to $1/C_t$ gives $K_{t+1} = \alpha\beta C_t / (1-\alpha\beta) = \alpha\beta A_t K_t^{\alpha}$. Taking logs and iterating backwards,

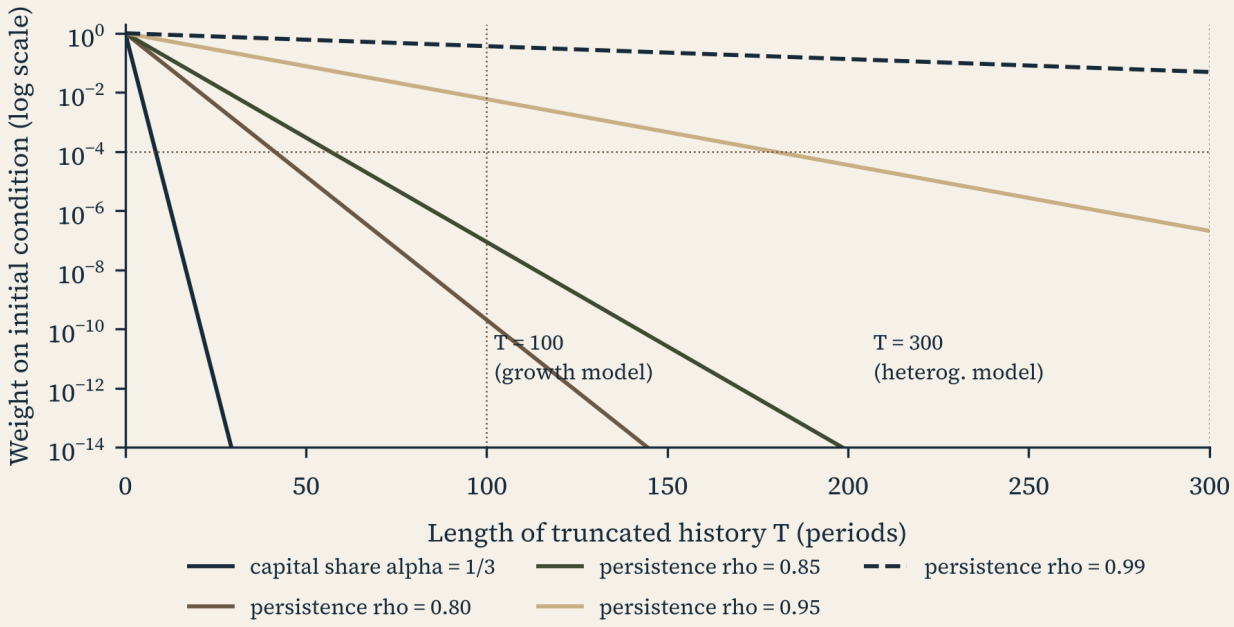
$$\log K_t = \underbrace{f(\alpha, \beta, A_{t-1}, \dots, A_{t-T})}_{\text{depends on the last } T \text{ shocks only}} + \underbrace{\alpha^T \log K_{t-T}}_{\text{truncation error}}. \tag{4}$$

Equation (4) is a geometric series: each backward step multiplies the inherited log capital by $\alpha < 1$, so the influence of the unobserved initial capital decays at rate α per period. The same argument for productivity, $\log A_t = g(\rho, \sigma, \varepsilon_{t-T}, \dots, \varepsilon_t) + \rho^T \log A_{t-T}$, shows that innovations are an equally good input when $|\rho| < 1$. The two statements combine in matrix form,

$$\begin{bmatrix} \log A_t \\ \log K_t \end{bmatrix} = \underbrace{\begin{bmatrix} \rho & 0 \\ 1 & \alpha \end{bmatrix}}_{M} \begin{bmatrix} \log A_{t-1} \\ \log K_{t-1} \end{bmatrix} + \begin{bmatrix} \sigma\varepsilon_t \\ \log(\alpha\beta) \end{bmatrix}, \tag{5}$$

where M is lower triangular with eigenvalues ρ and α. The memory of the initial condition therefore decays at the rate of the largest eigenvalue, $\max\{\rho, \alpha\}$. Figure 1 plots the decay weights and the history length needed for a given tolerance.

Figure 1. **Weight of the unobserved initial condition after T periods, for the contraction rates in the log-utility growth model**



Source: generated by the reviewer from equation (4), $\alpha = 1/3$; persistence values ρ are illustrative; dotted lines mark the $T = 100$ and $T = 300$ used in the paper and a 10^{-4} weight.

INFERENCE The figure makes the central limitation visible. For $\rho = 0.8$ a 10^{-4} weight needs $T \approx 41$ periods, for $\rho = 0.95$ about 180, and for $\rho = 0.99$ about 916 (computed as $\ln 10^{-4} / \ln \rho$). The paper itself notes that ρ close to one requires long histories (Section 3.2.2). At annual frequency with $\rho \approx 0.85$ the problem is mild, at quarterly frequency with persistent processes it is not.

3.2 Training objective

PAPER For a dataset $\mathcal{D}$ of state-history pairs, the loss is the mean squared relative Euler error,

$$\ell(\mathcal{D},\rho) = \frac{1}{N} \sum_{(\mathbf{X}_t^{seq,T},\mathbf{X}_t^{state})\in\mathcal{D}} \text{ree}\left(\mathbf{X}_t^{seq,T},\mathbf{X}_t^{state},\rho\right)^2, \tag{6}$$

with $K_{t+1} = \mathcal{N}_\rho(\mathbf{X}_t^{seq,T})$, $K_{t+2} = \mathcal{N}_\rho(\mathbf{X}_{t+1}^{seq,T})$ and the next history formed by dropping the oldest shock, $\mathbf{X}_{t+1}^{seq,T} = [(\mathbf{X}_t^{seq,T})_{2:T}, \varepsilon_{t+1}]$. The expectation inside ree is computed by Gauss-Hermite quadrature (eight nodes). The network input is exogenous, but the state vector $[A_t, K_t]$ must still be tracked in the simulation because resources $M_t = A_t K_t^\alpha L^{1-\alpha} + (1-\delta)K_t$, consumption $C_t = (1-s_t)M_t$ and $K_{t+1} = s_t M_t$ are computed from it. The network outputs the savings rate s_t through a sigmoid, so $0 < s_t < 1$ guarantees $C_t > 0$ and the budget constraint. Parameters are updated by Adam (Kingma and Ba 2015, reviewed elsewhere in this library). The paper prints the update as $\rho^{new} = \rho^{old} + \alpha^{learn}\nabla\ell$ (Equation 19), which as written is gradient ascent and should carry a minus sign; this is a typographical issue, not a conceptual one.

3.3 Overlapping generations economy with disasters

PAPER The second model has $H = 72$ cohorts with Cobb-Douglas production (Equation 25), capital and a bond, a convex capital adjustment cost $\psi(k',k) = \xi^{adj}(k'-k)^2$, short-sale limits, and stochastic depreciation driven by a two-regime Markov chain,

$$\delta_{t+1} = \delta\,\frac{2}{1+D_{t+1}}, \qquad \log D_{t+1} = \begin{cases} \rho_\delta \log D_t & \text{normal times,} \\ \mu_\delta + \rho_\delta \log D_t + \sigma_\delta \varepsilon_{t+1}^\delta & \text{disaster.} \end{cases} \tag{7}$$

With $\rho_\delta = 0$ and $D = 1$ in normal times, $\delta_{t+1} = \delta$. In disasters $\log D = \mu_\delta = -1.10$ implies $\delta_{t+1} \approx \delta \cdot 2/(1+e^{-1.1}) \approx 1.5\,\delta$, the 50 percent increase quoted in the paper, with substantial dispersion from $\sigma_\delta = 0.2$. The household of age h satisfies the Bellman equation $V_t^h = \max u(c_t^h) + \beta\mathbb{E}[V_{t+1}^{h+1}]$ with first-order conditions for the bond and capital,

$$p_t u'(c_t^h) = \beta\,\mathbb{E}\left[u'(c_{t+1}^{h+1})\right] + \lambda_t^{b,h}, \tag{8}$$

$$\left(1+\partial_{k'}\psi\right)u'(c_t^h) = \beta\,\mathbb{E}\left[u'(c_{t+1}^{h+1})\left(1-\delta_{t+1}+r_{t+1}^K - \partial_k\psi_{t+1}\right)\right] + \lambda_t^{k,h}, \tag{9}$$

with complementary slackness $0 = \lambda_t^h(x_{t+1}^{h+1} - \underline{x})$, $\lambda \geq 0$ and $x \geq \underline{x}$. Equation (8) prices the risk-free bond p_t through the marginal rate of substitution; Equation (9) says that the marginal cost of capital, including the adjustment cost, equals the discounted marginal benefit, which includes the stochastic return $1-\delta_{t+1}+r_{t+1}^K$. The inequality system is collapsed into one equation per age and asset with the Fischer-Burmeister function, $\psi^{FB}(a,b) = \sqrt{a^2+b^2} - a - b$, which equals zero if and only if $a \geq 0$, $b \geq 0$ and $ab = 0$ (Jiang 1999). The network input stacks $T = 144$ regime indicators and T innovations for each of the TFP and depreciation shocks, so $\mathbf{X}^{seq,T} \in \mathbb{R}^{3T} = \mathbb{R}^{432}$, twice the lifetime of a cohort.

INFERENCE The choice $T = 2H$ is plausible because a household's wealth depends on shocks over its own life, and prices depend on the wealth of other cohorts which themselves depend on earlier shocks. The paper gives no test of whether $T = 144$ is necessary or sufficient.

3.4 Heterogeneous households and heterogeneous firms

PAPER Firms use decreasing returns, $y_t^i = A_t z_t^i (k_t^i)^\alpha (l_t^i)^{1-\alpha-\zeta}$ (Equation 40), face asymmetric, smoothed adjustment costs $\psi(k',k) = \frac{\xi(k',k)}{2k}(k'/k-1)^2$ with $\xi_{down} > \xi_{up}$, and a non-negative dividend constraint. Aggregate volatility and the dispersion of idiosyncratic productivity move together in a two-state uncertainty regime, as in Bloom et al. (2018). The pure uncertainty shock is built by mean-preserving spreads of the three-state transition matrix, $\Pi_H^z = \Pi_L^z + u\,[\,1,-2,1\,]$ applied to each row's outer columns and centre, which leaves $\mathbb{E}[z'|z]$ unchanged. The static labour demand condition fixes the wage. Using $L=1$ and integrating labour demand over firms:

$$w_t = (1-\alpha-\zeta) A_t z_t^i (k_t^i)^\alpha (l_t^i)^{-\alpha-\zeta} \iff l_t^i = \left(\frac{A_t z_t^i (1-\alpha-\zeta)}{w_t}(k_t^i)^\alpha\right)^{\frac{1}{\alpha+\zeta}},$$
$$1 = \int l_t^i \, d\mu_t^F = \left(\frac{A_t(1-\alpha-\zeta)}{w_t}\right)^{\frac{1}{\alpha+\zeta}} \int \left(z^i (k^i)^\alpha\right)^{\frac{1}{\alpha+\zeta}} d\mu_t^F \implies w_t = A_t(1-\alpha-\zeta)\left(\int \left(z^i (k^i)^\alpha\right)^{\frac{1}{\alpha+\zeta}} d\mu_t^F\right)^{\alpha+\zeta}. \tag{10}$$

Equation (10) shows exactly how the firm distribution μ_t^F enters households' problems: the wage is a power mean of firm productivity weighted capital. This is the channel by which the cross-sectional distribution matters for aggregate outcomes and why the state must contain it or its sequence-space proxy.

The firm Euler condition with the multiplier λ_t^i on the dividend constraint is

$$\lambda_t^i = \frac{\Gamma\,\mathbb{E}\left[\Lambda_{t+1,t}\left(1 + r_{t+1}^{i,k} + \partial_{k'}\psi(k_{t+2}^i, k_{t+1}^i)\right)\left(1+\lambda_{t+1}^i\right)\right]}{1 + \partial_{k'}\psi(k_{t+1}^i, k_t^i)} - 1, \tag{11}$$

where Γ is the survival probability and $\Lambda_{t+1,t}$ the aggregate stochastic discount factor, defined as the shareholding-weighted average of household discount factors $\beta\,\mathbb{E}_e[u'(c')]/u'(c)$ (Equations 71 and 72). Households hold shares, with budget $c_t^i = e_t^i w_t + \theta_t^i(D_t + \Gamma p_t) - \theta_{t+1}^i p_t + \Pi_t^{SU}$ and first-order condition $p_t u'(c_t^i) = \beta\mathbb{E}[u'(c_{t+1}^i)(D_{t+1} + \Gamma p_{t+1})] + \lambda_t^i$.

3.5 Operator learning and shape preservation

PAPER Operator learning writes $f(X,x) = g(X)(x)$ where X is the aggregate state and x the idiosyncratic state. Here g is a network mapping the shock history to the values of the policy on a fixed grid $\Theta^{grid} = [\theta_0 < \cdots < \theta_N]$, and piecewise linear interpolation supplies off-grid values. For monotonicity, the network predicts positive increments and the consumption grid is their cumulative sum:

$$c_{t,i,j} = c_{t,i,0} + \sum_{k=1}^{j} dc_{t,i,k}, \qquad dc_{t,i,k} > 0. \tag{12}$$

For concavity as well, the network predicts $ddc < 0$ and the slope is built as a decreasing positive sequence,

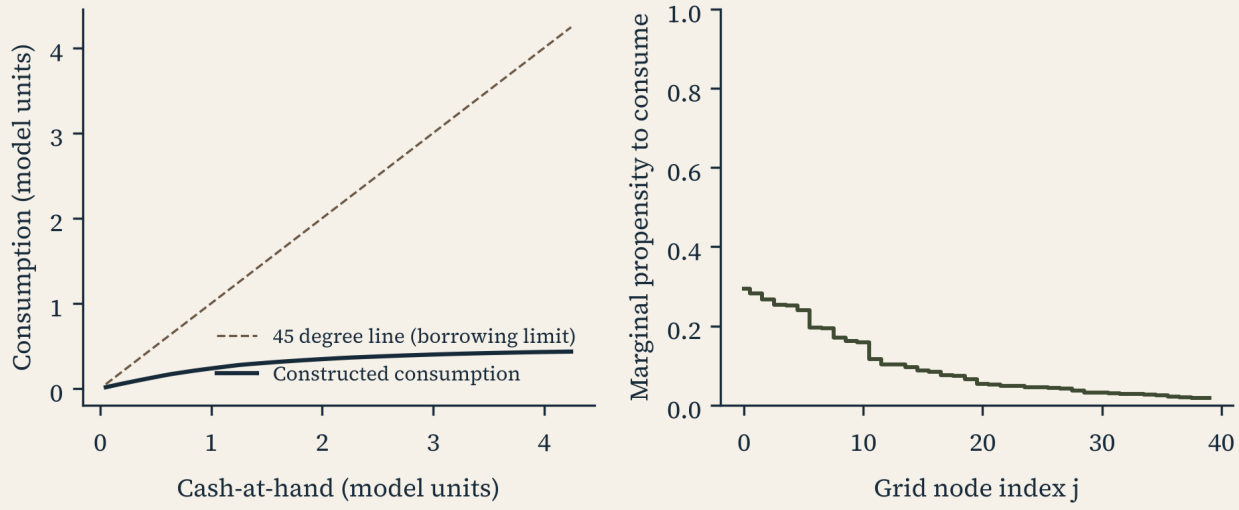
$$dc_{t,i,j} = d\theta_j \exp\Big(\sum_{k=1}^{j} ddc_{t,i,k} \Big), \tag{13}$$

so that $dc/d\theta_j = \exp(\sum_k ddc_k)$ is positive and decreasing in j. For the borrowing constraint the object predicted is the marginal propensity to consume out of cash-at-hand, $mpc_{t,i,j} = cs_{t,i,0} \exp(\sum_{k=1}^{j} dmpc_{t,i,k})$ with $cs_{t,i,0} \in (0, 1]$ from a sigmoid and $dmpc < 0$ from a negative softplus, hence

$$c_{t,i,0} = cs_{t,i,0}\, cah_{t,i,0}, \qquad c_{t,i,j} = c_{t,i,0} + \sum_{k=1}^{j} mpc_{t,i,k}\, dcah_{t,i,k}, \qquad dcah_{t,i,k} = d\theta_{t,i,k}(D_t + \Gamma p_t). \tag{14}$$

Equation (14) has three properties by construction: $0 < mpc \leq 1$, which makes $c \leq cah$ at every node (the borrowing limit $\theta' \geq 0$ holds because savings $cah - c$ are non-negative); mpc decreasing in j, which is concavity; and $mpc > 0$, which is monotonicity. Figure 2 illustrates the construction with arbitrary raw network outputs.

Figure 2. Shape-preserving construction of a consumption function from unconstrained outputs



Source: generated by the reviewer by applying equations (13) and (14) to random draws; model units are arbitrary and no paper data are used.

CRITIQUE Piecewise linear interpolation preserves monotonicity and concavity in one dimension only. The authors note that concavity is not generally preserved by multivariate linear interpolation, which matters if policies depend on two continuous idiosyncratic states.

3.6 Endogenous grid targets and market clearing

PAPER Because the consumption function is monotone, the endogenous grid method (Carroll 2006) inverts the Euler condition: given c_{t+1} from the current guess, next-period marginal utility fixes c_t on an endogenous asset grid without root-finding. The market-clearing price p_t is then found by Newton-Raphson on excess demand,

$$p_t^{new} = p_t^{old} - \frac{ED(p_t^{old}, \mathcal{A}_t, \mathcal{N}_\rho)}{\partial ED(p_t^{old}, \mathcal{A}_t, \mathcal{N}_\rho)/\partial p_t^{old}}, \qquad ED = \Theta_{t+1}^{guess} - 1. \tag{15}$$

The solved consumption grid $\mathcal{C}_t^{grid,solved}$ and price p_t^{solved} become targets, and the household and price losses are relative squared distances to targets (Equations 107 to 109), added to the firm KKT residuals in the weighted loss (Equation 110) with five weights.

INFERENCE Because the targets are produced from the network's own continuation values at $t+1$, the household block is trained as a fixed-point iteration of time iteration type rather than as direct minimisation of a global residual. Small supervised losses certify consistency of the iteration, not convergence to a unique equilibrium.

4 Data and Empirical Strategy

The paper has no empirical data. The experimental design is a computational one: calibrations are borrowed from the literature, training sets are simulated, and evaluation uses residuals on fresh simulated paths. The identifying question is therefore not causal but one of numerical validity: where does the evidence that the approximation is correct come from?

4.1 Calibration and simulated data

Parameters are taken from standard sources: Auclert et al. (2021) for the household income process ($\rho_e = 0.871$, $\sigma_e = 0.246$, discretised by the Rouwenhorst 1995 method to $e \in \{0.538, 1.462\}$), and Bloom et al. (2018) and Khan and Thomas (2008) for the uncertainty regime and the productivity process. Table 1 summarises the three applications as reported.

Table 1. **The three applications as reported in the paper**

Feature	Stochastic growth	OLG with disasters	Households and firms
Aggregate shocks	TFP AR(1), $\rho_A = 0.8$, $\sigma_A = 0.03$	TFP, depreciation, two-regime chain	TFP with two-regime volatility
History length T	100	144	300 per shock
Network input dimension	100	432	600
Hidden layers	3 of 128 (gelu)	3 of 720 (gelu)	3 of 1024 (gelu)
States per episode, mini-batch	4096, 256	8192, 64	131072, 128
Learning rate	10^{-5}	10^{-5}	10^{-6}
Training episodes	40000 (Table 2)	100352 in four stages	17100 in five stages

Source: Tables 1 to 6 of the paper (official); episode totals for the second and third models are sums of the stage counts computed by the reviewer (generated). The text of Figure 1 refers to 50000 episodes, whereas Table 2 states 40000.

4.2 Empirical specification and identification

EVIDENCE The identifying variation for accuracy is the dispersion of the residual across states drawn from the ergodic set of the economy, and in the first model the difference between the network policy and a policy obtained by policy time iteration on a dense grid (Judd 1998). In the second and third models no reference solution is available. In the third, the paper states that a closed form or high-quality numerical reference is not available, so accuracy rests on equilibrium conditions and on the comparison of network outputs with the EGM-Newton targets.

CRITIQUE Residual-based accuracy is a necessary condition. It identifies the policy only if the equilibrium conditions have a locally unique solution and if the sampled states cover the relevant region. Errors in the first model are of the same magnitude as the policy error (mean below 0.005 percent each), but the authors correctly say this proportionality cannot be generalised to other models.

4.3 Data limitations and biases

- Selection of test states: errors are evaluated on 4096 or 1024 states drawn from the same ergodic distribution used for training. Out-of-distribution histories, such as long runs of adverse shocks, are not tested.
- Look-ahead is not an issue because histories contain only past shocks, but the simulation also tracks the endogenous state, which is a function of the current network. The residual is therefore evaluated on the distribution induced by the policy being tested.
- Hyperparameter search is not described. Learning rates, widths, stage lengths and the weights w differ across models and may have been tuned on the evaluation metric. This is a form of mild data mining on the benchmark.
- No random seeds, run times, hardware or repeated-run variability are reported in the sections read.

5 Results

5.1 Stochastic growth model

EVIDENCE The loss falls steadily to well below 10^{-8} (Figure 1 of the paper). On the simulated ergodic set the mean absolute relative Euler error is below 0.005 percent and the 99.9th percentile is below 0.025 percent. Against the grid solution the mean relative policy difference is below 0.005 percent and the 99.9th percentile below 0.015 percent. In economic terms, an error of 0.005 percent in consumption means that an agent following the network policy would, on average, mis-set consumption by five cents per 1000 units, far below any calibration uncertainty. Statistical significance is not applicable: the experiment is deterministic given the network, and no confidence intervals over training runs are given. The result supports H1 and H2 in a case where the answer was known in advance, so it is a proof of concept rather than evidence for the harder claims, as the authors acknowledge.

5.2 Overlapping generations with disasters

EVIDENCE The mean and 90th percentile errors are below 0.1 percent and the 99.9th percentile below 0.32 percent for both assets and all ages; the mean error is about 0.03 percent. The model reproduces a life-cycle pattern: young households are constrained, savings accumulate until the labour endowment falls after age 62, and consumption in disasters falls most for households around age 80, who hold capital as their main asset. The age-80 result follows the paper's explanation that capital is predominantly held by older households. Economically the result illustrates why long histories are needed, since the disaster hits wealth built over decades.

5.3 Heterogeneous households and firms

EVIDENCE Firm Euler errors have mean below 0.07 percent, 90th percentile below 0.16 percent and 99.9th percentile below 0.4 percent, higher around kinks from the dividend constraint and asymmetric adjustment costs. The remaining KKT condition errors have mean below 0.2 percent and 99.9th percentile below 0.8 percent. Household consumption errors have mean below 0.08 percent, 90th percentile below 0.17 percent and 99.9th percentile below 0.4 percent. The price network deviates from the market-clearing price with mean 0.17 percent, 90th percentile 0.35 percent and 99.9th percentile 0.80 percent, while implied asset demand error is below 0.1 percent. The price block is the least accurate and matters most for asset pricing applications: a 0.17 percent mean price error is large relative to the size of an equity risk premium of a few percent per year, if the same function were used to compute returns. This is an inference; the paper reports no asset-pricing moments.

5.4 Tables analysed

EVIDENCE Table 2 reproduces the accuracy statistics. The question answered is how large the equilibrium residual is; the measured variable is the relative error in units of consumption or marginal cost; the columns show the distribution across sampled states. The key number is the 99.9th percentile, since it records near worst-case behaviour. The table fails to show out-of-distribution accuracy, accuracy conditional on rare regimes, or variation across training seeds.

Table 2. **Reported accuracy of the equilibrium residuals (percent of relative error)**

Model	Condition	Mean	90th pct	99.9th pct
Stochastic growth	Euler equation	0.005	n.r.	0.025
Stochastic growth	Policy versus grid solution	0.005	n.r.	0.015
OLG with disasters	Capital and bond optimality	0.03	0.10	0.32
Households and firms	Firm Euler (marginal cost)	0.07	0.16	0.40
Households and firms	Firm remaining KKT	0.20	n.r.	0.80
Households and firms	Household consumption	0.08	0.17	0.40
Households and firms	Equity price versus solved price	0.17	0.35	0.80

Source: Sections 3.6, 4.5 and 5.5 of the paper (official). The figures are upper bounds ("below") in the text for most rows; n.r. means not reported in the text read. For the OLG row the paper states mean and 90th percentile below 0.1 percent, with a mean around 0.03 percent.

5.5 Figures analysed

EVIDENCE The left panel of the paper's Figure 2 plots the distribution of absolute Euler errors with markers at the mean, 99th and 99.9th percentiles; the middle panel overlays the network policy on the grid solution and the right panel shows policy errors. The relationship shown is agreement of two numerical solutions, not a causal relationship. Nonlinearity is not an issue in this model. The paper's Figure 9 (consumption and marginal propensity to consume across aggregate states) supports the mechanism: for every aggregate state shown the consumption function is increasing and concave and the marginal propensity to consume lies between zero and one, but these are guaranteed by the architecture and therefore cannot count as evidence of accuracy. The firm dividend policy in Figure 8 shows two kinks, where the dividend constraint stops binding and where firms switch from investing to disinvesting. These are economically meaningful; the larger residuals near them suggest the network smooths kinks, a known limitation of smooth activations.

6 Robustness and Additional Tests

Table 3. Robustness tests: concern, test, result and assessment

Concern	Test reported	Result	Does it resolve the concern?
Truncated history is not sufficient	Closed form for $\delta = 1$, log utility; eigenvalue argument	Error decays as $\max\{\rho, \alpha\}^T$	Partly. Holds in one special case, and for other models it is assumed from ergodicity
Accuracy of the approximation	Comparison with grid-based policy time iteration in the growth model	Differences below 0.015 percent at the 99.9th percentile	Yes for that model, but not informative for the harder models
Implicit residual may not reflect policy error	Compare implicit and explicit errors in the growth model	Same order of magnitude	Partly. The authors say it is model dependent
Training instability	Step-wise homotopy: bond supply, adjustment costs, discount factor, short-lived asset parameter τ	Training converges in each model	No test of sensitivity to the schedule
Overfitting	Fresh simulated dataset each episode	Argued not to be a concern	Yes for sampling error, not for distribution shift
Kinks and constraints	Fischer-Burmeister formulation and inspection of residuals by state	Errors slightly larger near kinks	Partly. Tail errors reported but not decomposed
Alternative methods	None	Qualitative comparison with Lee (2025) and Auclert et al. (2021)	No

Source: reviewer's assessment of Sections 3 to 5 of the paper (generated).

CRITIQUE The paper has very few formal robustness exercises. There is no sensitivity to T, to network width, to random seeds, to the number of quadrature nodes, or to the weights in the loss. The essential robustness check for a sequence-space method, accuracy as a function of T, is missing and would directly test H1 in the harder models.

7 Critical Evaluation

7.1 What the paper establishes

ESTABLISHED

In the Brock and Mirman economy the method reproduces a high-accuracy grid solution to within 0.015 percent at the 99.9th percentile. The truncation error is exactly geometric in the log-utility, full-depreciation case, with decay rate $\max\{\rho, \alpha\}$. Monotone, concave, borrowing-constrained consumption grids can be generated by construction using Equations (12) to (14).

SUPPORTED BUT CONDITIONAL

Small equilibrium-condition residuals in the OLG and heterogeneous firm models, conditional on the sampled ergodic set and on the calibrated persistence being moderate ($\rho \leq 0.87$). The approach is conditional on the models being ergodic with fast-decaying memory.

SUGGESTIVE

That the sequence-space formulation improves training stability relative to state-space DEQN, because the input distribution does not move during training. The argument is plausible but no comparison is shown.

NOT ESTABLISHED

That the computed object is the unique equilibrium of the third model; that the approximation is accurate off the ergodic set or in rare histories; that the method is faster or more accurate than competing global methods; that truncation error is negligible in the third model; and any economic conclusion about distributional or asset pricing effects, none of which is reported.

7.2 Strengths

STRENGTHS RECOGNISED BY THE AUTHORS

- Generality across discrete and continuous shocks, regimes and stochastic volatility, which matters because non-Gaussian and regime shocks are the settings where local methods fail.
- Parallelisation on modern hardware, because trajectories can be simulated and trained simultaneously, which matters for scaling.
- Policies satisfy monotonicity, concavity and the borrowing limit by construction, which removes a class of spurious solutions.

STRENGTHS IDENTIFIED INDEPENDENTLY

- Transparent diagnostic: the decay-rate analysis in Equations (4) and (5) gives a rule for choosing T from model primitives, which is rare in deep-learning solution papers.
- Residuals are expressed in economically interpretable units (relative consumption and marginal cost errors), allowing comparison of accuracy across models.
- Using the endogenous grid method to create supervised targets replaces differentiating through a long computational graph, a numerically sensible design that reduces gradient noise.
- The three applications escalate in difficulty, with the first having a reference solution, which gives a credible chain of validation from known to unknown.

7.3 Weaknesses and critique

SINGLE MOST IMPORTANT WEAKNESS

Problem. Sufficiency of the truncated history is proven only for $\partial = 1$ with log utility, and the accuracy of the heterogeneous model is certified solely by residuals in the same simulated distribution, without an external benchmark. **Why it matters.** Residual accuracy proves that the equilibrium equations hold at sampled points, not that the sampled points cover the states that matter, or that the memory of the initial wealth distribution has vanished at $T = 300$. In heterogeneous agent models the wealth of the richest and least constrained households is highly persistent, so the contraction of the distribution can be much slower than the contraction of aggregate capital. **Potential bias.** A truncation bias that is invisible in the residual, because the residual conditions on the network's own history input and does not know what information is missing. **How it could be tested.** Train with increasing T and report residuals and aggregate moments as a function of T; compare with Reiter (2009) or the sequence-space Jacobian solution in the small-shock limit, where both methods must agree.

OTHER WEAKNESSES

- No baseline comparison. The claims relative to Lee (2025), Chien et al. (2011) and state-space DEQN are qualitative. Test: run state-space DEQN on the OLG model with the same budget.
- Tuning and reproducibility. The step-wise training uses at least four interacting homotopies in the third model (adjustment costs, the parameter τ, the discount factor weight, and sample size), with hand-chosen schedules. Test: random schedules and seeds with reported variation.
- Interpretation of residual levels. A mean error of 0.17 percent in the price is not small relative to equity premia. Test: compute the implied risk premium and its error.
- Generalisation. Fresh data each episode removes sampling overfitting but not evaluation on atypical histories. Test: evaluate on histories of stress, such as ten consecutive negative TFP shocks or long high-uncertainty spells.
- Economic content. No impulse responses, moments or welfare numbers are shown, so the paper does not demonstrate that the new solutions change any economic conclusion.

– Presentation. Gradient descent with a plus sign (Equation 19), the Fischer-Burmeister function is defined with opposite sign in the text and in Equation (97), the statement of ξ_{up} twice in Step 4 of Section 5.4.1, inconsistency between 40000 and 50000 training episodes, and a reference to Figure 9 for the dividend policy that appears in Figure 8.

8 Contradictions and Edge Cases

CRITIQUE The following conditions are explored by direct reasoning and by calculation on the formulae of the paper.

Persistence near a unit root. The weight on the initial condition is $\max\{\rho, \alpha\}^T$. To reach 10^{-4} the history must have length $\ln 10^{-4} / \ln \rho$, which is 8 for $\alpha = 1/3$, 41 for $\rho = 0.8$, 57 for $\rho = 0.85$, 180 for $\rho = 0.95$ and 916 for $\rho = 0.99$ (generated, Figure 1). For a quarterly model with $\rho = 0.99$ and three shocks the input would have more than 2700 dimensions. For $\rho = 1$ (permanent shocks such as growth rates) the truncation error never vanishes, and the method needs detrending or a different representation.

Repeated eigenvalues. If $\rho = \alpha$, the matrix M in Equation (5) has a Jordan block and the influence of the initial condition decays as $T\rho^T$, not ρ^T. The result stated in the paper (rate $\max\{\rho, \alpha\}$) is exact only for distinct eigenvalues; in the neighbourhood of $\rho = \alpha$ the required T is larger by a factor that grows with T.

Heterogeneity changes the slowest eigenvalue. In the full-depreciation model the slowest mode is α or ρ. In an Aiyagari or Krusell and Smith economy, households at the borrowing constraint forget their wealth quickly, but a household with large wealth and a low marginal propensity to consume out of income keeps its wealth for many periods. The dominant decay rate of the wealth distribution is therefore likely to be higher than α or ρ, so $T = 300$ at annual frequency, which is generous, may still be close to a limit when the model is recalibrated to quarterly frequency (Inference).

Accuracy in the tails. The mean residual hides the tails, but even the 99.9th percentile is evaluated on typical draws. A history with ten consecutive adverse shocks has probability of the order $0.16^{10} \approx 10^{-8}$ under a standard normal tail at -1 and is almost never sampled in training, so the network may extrapolate poorly in exactly the states where policy relevance (crises) is highest (illustrative calculation).

Scale of the approximators. For the third model the four networks have approximately 3.3, 3.3, 2.9 and 2.7 million weights, about 12 million in total, assuming dense layers as in Table 6 and ignoring biases (estimate based on the layer widths stated in the paper). The training set of $17100 \times 131072 \approx 2.2 \times 10^9$ state-history pairs is sizeable, and a method requiring this volume of simulation should be compared in cost with a Reiter-type solution of the same model.

Alternative mechanisms for small residuals. A network could produce small residuals by exploiting the simulated states' concentration near the steady state; if shocks are small ($\sigma_A = 0.03$ and 0.0124), the economy stays close to the steady state, and the global nature of the solution is lightly tested. A comparison with the first-order solution in the same region would show how much nonlinearity is actually captured. In the disaster model $\mu_\delta = -1.10$ and the 1.5 factor produce a genuine nonlinearity, which is a good test.

External validity. Models are calibrated annually with Cobb-Douglas production, CRRA utility and two-state idiosyncratic income. Richer income processes would multiply the grid in the operator network ($2 \times N$ outputs per aggregate state); applications with many idiosyncratic states would need larger outputs.

9 Further Literature

Table 4. **Literature connections**

Paper	Shared idea	Relationship	Why it matters
Krusell and Smith (1998)	Approximate the distribution by a low-dimensional statistic	Foundation; the present paper replaces moments by shock histories	Defines the problem and the state-space benchmark
Chien, Cole and Lustig (2011)	Summarise the state by past aggregate shocks	Direct predecessor, discrete shocks, limited persistence	Origin of the sequence-space idea
Lee (2025)	Repeated transition method over a simulated path	Competing global method, strong for few discrete shocks	The closest alternative; no direct comparison
Azinovic, Gaegauf and Scheidegger (2022)	Deep equilibrium nets trained on residuals along simulated paths	Training framework extended here	Source of loss, simulation and error metrics
Maliar, Maliar and Winant (2021)	Neural networks trained on Euler and Bellman residuals	Parallel development of the same family	Shows the approach is general
Auclert, Bardoczy, Rognlie and Straub (2021)	Sequence-space Jacobian, linearisation in aggregates	Local complement; fast but certainty equivalent	Natural benchmark in the small-shock limit
Reiter (2009)	Projection plus perturbation for aggregate risk	Local-in-aggregates method with global idiosyncratic part	Alternative validation for the third model
Carroll (2006)	Endogenous gridpoint method	Used to build supervised targets	Removes root-finding in the Euler step
Zhong (2023)	Operator learning in macroeconomics	Basis for the architecture; shape constraints are new here	Defines the network-to-function mapping
Bloom et al. (2018)	Uncertainty shocks with stochastic volatility	Source of the calibration and the economic question	Provides a benchmark for results in the third model
Kase, Melosi and Rottner (2023)	Solve over parameter ranges for estimation by likelihood	Possible downstream use	Suggests how the method could be estimated on data
Kingma and Ba (2015)	Adam optimiser	Used for training; reviewed in this library in the Optimization series	Learning-rate sensitivity relevant to the unstable stages

Source: relationships are assessed by the reviewer from the paper's own discussion of the literature (generated); bibliographic details are in the references.

10 Research Gaps

10.1 Paper-level gaps

- **Gap:** no accuracy-versus-T study. *Why it matters:* H1 is the core assumption. *Existing limitation:* one closed-form case. *Way forward:* train a sequence of networks with $T \in \{10, 25, 50, 100, 300\}$ and report residuals and moments.
- **Gap:** no economic output from the third model. *Why it matters:* the method is justified by its use. *Existing limitation:* only accuracy is reported. *Way forward:* impulse responses to an uncertainty shock, state-dependent multipliers, asset prices and risk premia.
- **Gap:** no out-of-distribution testing. *Why it matters:* crises are rare histories. *Existing limitation:* test states are drawn from the training distribution. *Way forward:* importance-sampled stress histories.

10.2 Literature-level gaps

- **Gap:** no common benchmark suite for global heterogeneous agent solvers. *Why it matters:* claims cannot be compared. *Existing limitation:* each paper uses its own calibration and accuracy metric. *Way forward:* a public set of models with high-quality reference solutions, accuracy, run time and memory reported on equal hardware.
- **Gap:** theory of when shock histories are sufficient statistics in models with heterogeneity. *Why it matters:* it decides T. *Existing limitation:* the argument rests on ergodicity. *Way forward:* spectral bounds on the contraction of the distribution map.

10.3 Methodological gaps

- **Gap:** estimation. *Why it matters:* a solver is useful if it can be taken to data. *Existing limitation:* Kase et al. (2023) estimate with parameters as inputs, but not in the sequence space. *Way forward:* amortised solutions with parameters and histories as inputs, then likelihood or simulated moments.
- **Gap:** certified error bounds. *Why it matters:* residuals are not errors. *Existing limitation:* no bound relating the Euler residual to the policy error outside simple models. *Way forward:* a posteriori bounds based on the contraction properties of time iteration.

11 Research Opportunities

Idea 1. Benchmarking the sequence-space solution against sequence-space Jacobians

RESEARCH QUESTION. How large is the gap between the global sequence-space neural solution and the local sequence-space Jacobian solution as the size of aggregate shocks grows, and where does the gap exceed numerical error?

MOTIVATION. The paper contrasts itself with local methods but does not measure the difference. Quantifying the wedge tells practitioners when global methods are worth their cost.

HYPOTHESIS. For shocks below about one standard deviation the two methods agree to within the neural residual; the discrepancy in second moments and risk premia grows quadratically in shock size.

ECONOMIC MECHANISM. Certainty equivalence in aggregates removes precautionary and risk-premium effects that the global solution retains.

LITERATURE CONNECTION. Auclert et al. (2021), Reiter (2009), Krusell and Smith (1998).

DATA. Simulated; a standard Krusell and Smith economy and the paper's third model calibrated as in Table 5.

VARIABLES. Policy functions, impulse responses, second moments, wealth Gini, price of risk.

METHOD. Solve both ways for shock volatility scaled from 0.25 to 4 times baseline and compare ergodic moments and impulse responses.

IDENTIFICATION STRATEGY. The linear solution is exact at first order, so any divergence at small shocks measures numerical error, and divergence at larger shocks measures nonlinearity.

EXPECTED CONTRIBUTION. A validation standard for global heterogeneous agent solvers and an estimate of the nonlinearity wedge.

MAIN RISK. Numerical error of the neural solution may exceed the wedge for plausible shock sizes.

Idea 2. Minimal history length and memory of the wealth distribution

RESEARCH QUESTION. What is the shortest history T^* that achieves a given accuracy as a function of model primitives such as the discount factor, income persistence and borrowing limit?

MOTIVATION. The choice of T is currently heuristic and drives cost.

HYPOTHESIS. T^* scales as $\ln \varepsilon / \ln \lambda_2$ where λ_2 is the slowest eigenvalue of the linearised distribution dynamics, and is larger than the aggregate rate $\max\{\rho, \alpha\}$ in heterogeneous models.

ECONOMIC MECHANISM. Rich households have low marginal propensities to consume and forget initial wealth slowly.

LITERATURE CONNECTION. Chien et al. (2011), Lee (2025), Auclert et al. (2021).

DATA. Simulated, and eigenvalues from the sequence-space Jacobian of the same calibration.

VARIABLES. Residual, T, λ_2, discount factor, income persistence.

METHOD. Train networks for a grid of T and primitives; regress log error on T; compare slope with $\ln \lambda_2$.

IDENTIFICATION STRATEGY. Vary primitives that change λ_2 with known direction while holding the architecture fixed.

EXPECTED CONTRIBUTION. A rule for choosing T and a test of H1 for heterogeneous economies.

MAIN RISK. Network capacity limits may dominate truncation error and mask the scaling.

Idea 3. Asset pricing implications of the uncertainty-regime economy

RESEARCH QUESTION. Does the third model, solved globally, generate a counter-cyclical equity premium and a volatility risk premium of the observed order when uncertainty shocks raise idiosyncratic dispersion?

MOTIVATION. The paper solves a model with an equity price but reports no asset-pricing moment, the natural use of a global solution.

HYPOTHESIS. Higher idiosyncratic uncertainty raises the shareholders' stochastic discount factor volatility through the constraint on dividends and the heterogeneity of the discount factor, increasing conditional premia.

ECONOMIC MECHANISM. Uninsurable income risk and dividend constraints transmit uncertainty to the price of equity (Equations 11 and the household KKT condition).

LITERATURE CONNECTION. Bloom et al. (2018), Shiller (1981) on excess volatility, Fama and French (1993) on factor premia, both reviewed in this library.

DATA. Model-simulated returns; empirical moments from CRSP and Shiller's data for comparison.

VARIABLES. Equity return, conditional Sharpe ratio, price-dividend ratio, regime indicator.

METHOD. Simulate long paths, compute conditional moments by regime, compare with data.

IDENTIFICATION STRATEGY. Comparative statics: set the idiosyncratic spread $u = 0$ to isolate the uncertainty channel.

EXPECTED CONTRIBUTION. First quantitative asset-pricing evaluation of a global solution of this model class.

MAIN RISK. Price network error of 0.17 percent may be comparable to the premium, requiring a more accurate price block.

Idea 4. Stress-history accuracy and importance sampling

RESEARCH QUESTION. Is accuracy preserved on rare adverse histories, and does importance sampling of training histories improve it?

MOTIVATION. Crises are the use case for global methods and the least sampled states.

HYPOTHESIS. Residuals rise by a factor of several on tail histories, and tilting the shock distribution during training reduces the tail error at little cost in the bulk.

ECONOMIC MECHANISM. Policies near binding constraints are strongly nonlinear and trained on few samples.

LITERATURE CONNECTION. Azinovic et al. (2022); extreme-value and importance sampling literature.

DATA. Simulated; stress histories constructed from the shock process.

VARIABLES. Residual by history percentile, loss weights, tilt parameter.

METHOD. Train with a tilted proposal and reweight the loss, then evaluate on a stress panel.

IDENTIFICATION STRATEGY. Compare tilted and untilted training on identical test panels.

EXPECTED CONTRIBUTION. A tail-accuracy protocol that complements mean residuals.

MAIN RISK. Importance weights may have high variance.

Idea 5. Social security and disaster risk in the overlapping generations economy

RESEARCH QUESTION. What pay-as-you-go pension design maximises welfare when disasters redistribute wealth across cohorts?

MOTIVATION. The OLG model shows that disasters hit older households most, so insurance across cohorts has value; the paper omits taxes and social security.

HYPOTHESIS. State-contingent pensions, indexed to the disaster regime, improve ex ante welfare relative to fixed benefits.

ECONOMIC MECHANISM. Incomplete markets prevent insurance against capital depreciation shocks; the pension is a risk-sharing device.

LITERATURE CONNECTION. Krueger and Kubler (2004) on OLG with stochastic production; Azinovic and Zemlicka (2024).

DATA. Disaster frequencies and sizes from the Barro and Ursua macroeconomic crisis data (not accessed in this review).

VARIABLES. Consumption by age and regime, pension replacement rate, tax rate, welfare in consumption equivalents.

METHOD. Extend the OLG model with a policy parameter as an additional network input and optimise welfare.

IDENTIFICATION STRATEGY. Comparative statics across pension rules in the same solved economy.

EXPECTED CONTRIBUTION. A tractable global analysis of intergenerational risk sharing.

MAIN RISK. Calibration of disaster parameters is weakly disciplined by data.

11.1 Ranking

Table 5. **Ranking of research ideas (scores from 1 to 5, assigned by the reviewer)**

Idea	Novelty	Feasib.	Data	Method	Import.	Contrib.	Total
1. Benchmark against Jacobians	3	4	5	4	4	4	24
2. Minimal history length	5	4	5	4	4	5	27
3. Asset pricing of the model	4	3	4	4	5	5	25
4. Stress-history accuracy	4	4	5	4	3	4	24
5. Pensions and disasters	4	3	3	3	4	4	21

Source: reviewer's subjective scoring.

The top three are Ideas 2, 3 and 1. Idea 2 is the most novel and directly tests the core assumption of the paper with only simulated data. Idea 3 has the highest economic importance for a finance reader and turns a solver into a result. Idea 1 is the cheapest and supplies the validation on which the other two rely.

12 Learning Framework

12.1 Prerequisite concepts

- Recursive competitive equilibrium and the Krusell and Smith problem, to understand why the distribution is part of the state.
- Euler equations, Karush-Kuhn-Tucker conditions and complementarity problems.
- Feed-forward neural networks, activation functions, stochastic gradient descent and Adam.
- Ergodicity and contraction of Markov processes, and eigenvalue analysis of linear systems.
- Gauss-Hermite quadrature and Rouwenhorst discretisation.

12.2 Core concepts learned

- The aggregate state can be replaced by a finite window of exogenous shocks with an exponentially small error.
- Predicting a function of the idiosyncratic state from the aggregate state (operator learning) allows constraints to be imposed exactly.
- A monotone policy enables the endogenous grid method and supervised learning of equilibrium objects.

12.3 Important equations

Equation (4) for truncation error, Equation (5) for the contraction rate, Equation (10) for the wage as a power mean over firms, Equations (12), (13) and (14) for shape-preserving construction, and Equation (15) for market clearing.

12.4 Methods learned

Truncated-history state representation, deep equilibrium training on simulated ergodic paths, Fischer-Burmeister reformulation of complementarity conditions, shape-preserving operator networks, endogenous grid targets with Newton-Raphson market clearing, and step-wise model homotopy.

12.5 Questions to test understanding

COMPREHENSION

1. Why is the distribution of truncated shock histories independent of the policy being trained, and why does this stabilise training?
2. How does the paper obtain a consumption function that is increasing, concave and respects the borrowing limit?
3. What does a relative consumption error of 0.1 percent mean?

ANALYSIS

1. Derive $K_{t+1} = \alpha\beta A_t K_t^{\alpha}$ and the truncation error in Equation (4).
2. Why does equality of eigenvalues $\rho = \alpha$ change the decay rate to $T\rho^T$?
3. Why is a small residual not sufficient to guarantee a small policy error?

RESEARCH

1. Design an experiment that measures the dominant memory length of a heterogeneous agent economy without a reference solution.
2. How would the method change for a model with a permanent shock to the level of productivity?
3. How could the network be trained to deliver confidence bounds on equilibrium objects?

12.6 Papers to read next

Krusell and Smith (1998); Auclert, Bardoczy, Rognlie and Straub (2021); Azinovic, Gaegauf and Scheidegger (2022); Maliar, Maliar and Winant (2021); Reiter (2009).

13 Conclusion

13.1 One-sentence thesis

A truncated history of exogenous shocks, together with a shape-preserving neural operator, can serve as a global approximation of the equilibrium of heterogeneous agent economies, with equilibrium residuals below a fraction of one percent in the three models solved.

13.2 Three most important insights

- Sufficiency of the history is a statement about contraction: the error is $\max\{\rho, \alpha\}^T$ in the benchmark and the required T explodes as persistence approaches one.
- Building monotonicity, concavity and the borrowing limit into the output layer turns a learning problem into a constrained-by-construction one, and enables classical numerical tools for the targets.
- Small residuals are evidence, not proof; the paper's strongest validation comes from the model where a reference exists.

13.3 Strongest evidence

The agreement between the network and the grid-based solution in the stochastic growth model, with 99.9th percentile policy differences below 0.015 percent.

13.4 Biggest weakness

The absence of an external benchmark and of an accuracy-versus-T study in the heterogeneous economies, where sufficiency is assumed rather than shown.

13.5 Most important research gap

A theory and a diagnostic for the minimal history length in heterogeneous agent models.

13.6 Best research opportunity

Use the third model as a laboratory for asset pricing under uncertainty shocks (Idea 3), after validating it against sequence-space Jacobians in the small-shock limit (Idea 1).

13.7 Three papers to read next

Auclert, Bardoczy, Rognlie and Straub (2021); Azinovic, Gaegauf and Scheidegger (2022); Chien, Cole and Lustig (2011).

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